

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030597

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Query 1: Create the database

```
CREATE DATABASE IF NOT EXISTS ddl_db;  
USE ddl_db;
```

Output: -

✓	39	15:09:29	CREATE DATABASE IF NOT EXISTS ddl_db	1 row(s) affected	0.110 sec
✓	40	15:09:34	USE ddl_db	0 row(s) affected	0.015 sec

Query 2: -DDL (DATA DEFINITION LANGUAGE)

CREATE TABLE

```
2520030597-Vidhath  
CREATE TABLE bank_transactions (  
    txn_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    branch_name VARCHAR(50),  
    transaction_type VARCHAR(20),  
    amount DECIMAL(10,2),  
    transaction_date DATE  
);
```

Output: -

✓	41	15:12:39	CREATE TABLE bank_transactions (txn_id INT P...	0 row(s) affected	0.203 sec
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Query 3: -ALTER TABLE ADD COLUMN

```
);  
ALTER TABLE bank_transactions  
ADD account_no VARCHAR(20);
```

Output: -

✓	42	15:14:01	ALTER TABLE bank_transactions ADD account_no ...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.093 sec
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Query 4: -ALTER TABLE MODIFY COLUMN

2520030597-Vidhath
6.16.00.0000A-R3W1C119
ALTER TABLE bank_transactions
MODIFY customer_name **VARCHAR**(100);

Output: -

✓	43	15:15:05	ALTER TABLE bank_transactions MODIFY customer...	0 row(s) affected	Records: 0 Duplicates: 0 Warnings: 0	0.109 sec
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Query 5: -RENAME TABLE

RENAME TABLE
bank_transactions **TO** customer_transactions;

Output: -

✓	44	15:17:35	RENAME TABLE bank_transactions TO customer_tr...	0 row(s) affected		0.063 sec
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Query 6: -TRUNCATE TABLE

TRUNCATE TABLE customer_transactions;

Output: -

✓	46	15:19:47	TRUNCATE TABLE customer_transactions	0 row(s) affected		0.032 sec
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Query 7: -DROP TABLE

2520030597-Vidhath
6.16.00.0000A-R3W1C119
DROP TABLE IF EXISTS bank_backup;

Output: -

⚠	48	15:21:33	DROP TABLE IF EXISTS bank_backup	0 row(s) affected, 1 warning(s): 1051 Unknown table '...		0.109 sec
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Query 8: - DML (DATA MANIPULATION LANGUAGE)

INSERT SAMPLE DATA

```

1 • CREATE TABLE customers_transactions (
2     txn_id INT PRIMARY KEY,
3     customer_name VARCHAR(100),
4     branch_name VARCHAR(50),
5     transaction_type VARCHAR(20),
6     amount DECIMAL(10,2),
7     transaction_date DATE
8 );
9 • INSERT INTO customers_transactions VALUES
0 (101, 'Vidhath', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
1 (102, 'Raghuveer', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
2 (103, 'Durga prasad', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
3 (104, 'Vishnu', 'Vizag', 'Deposit', 8000, '2024-01-10'),
4 (105, 'Yashwanth', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
5 (106, 'Adithya Vardhan', 'Vizag', 'Deposit', 15000, '2024-01-12'),
6 (107, 'Varshith Reddy', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
7 (108, 'Varnika Guptha', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
8 (109, 'Prerna', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
9 (110, 'Nihita', 'Vijayawada', 'Deposit', 11000, '2024-01-16');

```

Output: -

✓	52	15:27:20	CREATE TABLE customers_transactions (txn_id I...	0 row(s) affected	0.031 sec
✓	53	15:27:28	INSERT INTO customers_transactions VALUES (101...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.015 sec

Query 9: -INSERT

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```

INSERT INTO customers_transactions
VALUES
(111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');

```

Output: -

✓	54	15:28:57	INSERT INTO customers_transactions VALUES (111...	1 row(s) affected	0.062 sec
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Query 10: -UPDATE

```

UPDATE customers_transactions
SET amount = 5000
WHERE txn_id = 105;

```

Output: -

56 15:30:53 UPDATE customers_transactions SET amount = 500... 1 row(s) affected Rows matched: 1 Changed: 1 War... 0.015 sec

Query 11: -DELETE

```
DELETE FROM customers_transactions
WHERE txn_id = 111;
```

Output: -

58 15:32:09 DELETE FROM customers_transactions WHERE txn... 1 row(s) affected 0.000 sec

Query 12: -SELECT

```
SELECT *
FROM customers_transactions;
```

Output: -

60 15:33:32 SELECT * FROM customers_transactions LIMIT 0, 10... 10 row(s) returned 0.016 sec / 0.000 sec

Result Grid

Filter Rows:

Edit:

Export/Import:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Vidhath	Hyderabad	Deposit	5000.00	2024-01-05
	102	Raghuveer	Hyderabad	Withdrawal	2000.00	2024-01-06
	103	Durga prasad	Vijayawada	Deposit	12000.00	2024-01-08
	104	Vishnu	Vizag	Deposit	8000.00	2024-01-10
	105	Yashwanth	Hyderabad	Withdrawal	5000.00	2024-01-11
	106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12
	107	Varshith Reddy	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14
	109	Prerna	Vizag	Withdrawal	4000.00	2024-01-15
	110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16
✱	NULL	NULL	NULL	NULL	NULL	NULL

Query 13: -SELECT WITH WHERE

```
SELECT *
FROM customers_transactions
WHERE transaction_type='Deposit';
```

Output: -

62 15:35:53 SELECT * FROM customers_transactions WHERE tra... 6 row(s) returned 0.000 sec / 0.000 sec

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Vidhath	Hyderabad	Deposit	5000.00	2024-01-05
	103	Durga prasad	Vijayawada	Deposit	12000.00	2024-01-08
	104	Vishnu	Vizag	Deposit	8000.00	2024-01-10
	106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12
	108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14
	110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16
*	NULL	NULL	NULL	NULL	NULL	NULL

Query 14: -SELECT WITH ORDER BY

```
SELECT *
FROM customers_transactions
ORDER BY amount DESC;
```

Output: -

64

15:37:59

SELECT * FROM customers_transactions ORDE...

10 row(s) returned

0.016 sec / 0.000 sec

Result Grid

Filter Rows:

Edit:

Export/Import:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	106	Adithva Vardhan	Vizag	Deposit	15000.00	2024-01-12
	103	Durga prasad	Vijayawada	Deposit	12000.00	2024-01-08
	110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16
	108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14
	104	Vishnu	Vizag	Deposit	8000.00	2024-01-10
	101	Vidhath	Hyderabad	Deposit	5000.00	2024-01-05
	105	Yashwanth	Hyderabad	Withdrawal	5000.00	2024-01-11
	109	Prerna	Vizag	Withdrawal	4000.00	2024-01-15
	102	Raghuveer	Hyderabad	Withdrawal	2000.00	2024-01-06
	107	Varshith Reddy	Vijayawada	Withdrawal	1000.00	2024-01-13
*	NULL	NULL	NULL	NULL	NULL	NULL

Query 15: - DCL (DATA CONTROL LANGUAGE)

GRANT ALL

```
GRANT ALL PRIVILEGES
ON ddl_db.customers_transactions
TO 'BranchManager'@'localhost';
SHOW GRANTS FOR 'BranchManager'@'localhost';
```

Output: -

✓	72	15:48:57	GRANT ALL PRIVILEGES ON ddl_db.customers...	0 row(s) affected	0.000 sec
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Query 16: -REVOKE ALL

```
REVOKE ALL PRIVILEGES
ON ddl_db.customers_transactions
FROM 'BranchManager'@'localhost';
```

Output: -

✓	74	15:49:33	REVOKE ALL PRIVILEGES ON ddl_db.customer...	0 row(s) affected	0.000 sec
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Query 17: -TCL (TRANSACTION CONTROL LANGUAGE)

COMMIT

```
UPDATE customer_transactions
SET amount=6000
WHERE txn_id=101;

COMMIT;
```

Output: -

✓	76	15:52:18	UPDATE customer_transactions SET amount=60...	0 row(s) affected Rows matched: 0 Changed: 0 ...	0.000 sec
✓	77	15:52:25	COMMIT	0 row(s) affected	0.000 sec

Query 18: - SAVEPOINT

```
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-- 2520030597-Vidhath
SAVEPOINT Before_Update;
```

Output: -

✓	78	15:53:32	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
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Query 19: -ROLLBACK TO SAVEPOINT

2520030597-Vidhath

2520030001-Ramitha

```
UPDATE customer_transactions
```

```
SET amount=99999
```

```
WHERE txn_id=102;
```

```
ROLLBACK TO Before_Update;
```

Output: -

✓	84	15:55:57	UPDATE customer_transactions SET amount=99...	0 row(s) affected Rows matched: 0 Changed: 0 ...	0.000 sec
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